EPOCH bugs and improvements

Contributors to this register

Official `main` checked 30 October 2026

1 How to read this document

This is a shared list of bugs and small improvements to EPOCH. The official `main` branch is the reference point. Bug entries concern faults in official EPOCH, including ones fixed only in a fork; defects in fork-only features are excluded. For bugs, the colour shows where a fix is available: **Fix: main** in official `main`, **Fix: 4.20-devel** in an official development branch, **Fix: fork** in a separate copy of EPOCH, or **Not fixed** if no fix is known. The yellow flag can name any official development branch. A purple flag does not mean the fix has reached official EPOCH. Click a purple flag to open the fork or branch named by that row. Credits recognise contributors; the detailed entries attribute particular fixes.

For improvements, the colour shows where the improvement is available: **Main**, **4.20-devel**, **Fork**, or **Not available**. The branch checked for this edition of the register is `f294c484f`. The tables are short; click an ID for the explanation and source links. These checks compare EPOCH files and commit records. They are not new EPOCH runs.

2 Adding an entry

Add one short row to `registry.tex` or `improvements_registry.tex`, then add a matching entry to `items.tex` or `improvements.tex`. Use a subsection for a serious bug and smaller headings for the failure, fix and test. Record the official branch checked and the date when changing a flag. The copyable examples are in `CONTRIBUTING.md`.

3 Bugs

Bug list

Bug IDs are numbers. Keep each description short; the middle column wraps when needed. Long paths and test results belong below the table.

ID	Bug	Fix status	Credits
1	Laser absorption diagnostic at maximum boundaries	**Fix: main**	Zheyuan Chen
2	2D linear Breit–Wheeler selects the wrong species	**Fix: main**	Zheyuan Chen
3	Photon probes can deadlock with particle IDs	**Fix: 4.20-devel**	Tom Goffrey
4	Particle IDs can be reused after restart or file loading	**Fix: fork**	Zheyuan Chen
5	Higuera–Cary particle push uses the wrong field scale	**Fix: 4.21-devel**	Zheyuan Chen
6	Trident can produce pairs when pair production is disabled	**Fix: fork**	Zheyuan Chen
8	`identify` can silently replace charge or mass	**Fix: fork**	Zheyuan Chen and Diana Gorlova

1. Laser absorption at maximum boundaries

Energy injected at `x_max`, `y_max` or `z_max` could be recorded with the wrong sign. This could make the reported absorption coefficient zero. The fix covers 1D, 2D and 3D laser code. Tom Goffrey merged PR #896 on 10 April 2026 as `0984b1d31`. It is in official `main` and in the v4.20.1 release notes (issue #893).

2. Linear Breit–Wheeler species selection in 2D

A one-line typo in the 2D linear Breit–Wheeler routine selected the wrong species. PR #892 merged `ba04ae645` on 26 March 2026. The fix is in official `main` and listed in v4.20.1 (issue #891). This is separate from the nonlinear Breit–Wheeler items below.

3. Photon-probe particle-ID deadlock

When it happens

With photon probes and `PARTICLE_ID` or `PARTICLE_ID4`, a rank with no photons could return before a call that all ranks needed to make. Other ranks would wait indefinitely.

Where the fix is

PR #906 merged `241d6fb9c` into the official `4.20-devel` branch on 1 June 2026, in all three dimensions. It is not in the official `main` version checked for this document. Zheyuan Chen made the later 2D fork commit `3e4734522` without knowing the fix was already in the official development branch. That commit also includes a separate ID fix (bug 4).

Test to keep

Run with at least two MPI ranks, a photon probe, and one rank without photons. Check that the run finishes and that the IDs remain unique. One rank cannot expose this deadlock.

4. Particle IDs reused after restart or file loading

Problem. The 2D investigation found that the largest assigned ID was not always carried through restart or particle loading from a file. New particles could then reuse an old ID.

Fix and status. `3e4734522` saves and raises the ID counter in the fork. The checked official `main` does not contain those changes. The fork fix is 2D only. A regression test should check IDs after particles leave the domain, after a restart with a different MPI layout, and after loading both supplied and automatic file IDs.

5. Higuera–Cary particle-push field scale

Problem

The magnetic rotation in the Higuera–Cary particle push uses an interpolated field. Its scale must include the same factor used elsewhere in the push. In the checked official `main`, that factor is missing from this rotation.

Fix and test

Zheyuan Chen's fix `2684956bf` changes the calculation in 1D, 2D and 3D. Tom Goffrey merged it into official `4.21-devel` through PR #925 on 6 October 2026 (merge `2b55e10cf`). The same correction was also made in the fork as `d86ceceab`. It is absent from the checked official `main`. A useful test is a single particle in crossed fields, compared with an independent trajectory, for several particle shapes. This document checks the source; it does not rerun that test on the official build.

6. Trident pair creation with pairs disabled

Problem

With `TRIDENT_PHOTONS` compiled and `produce_pairs = F` in the deck, the electron loop could still try to create a trident pair. If no pair species was set, this could use species index `-1` and crash.

Fix and test

The fork commit `1a88d4afa` checks `produce_pairs` before changing the trident event counter and before creating a pair, in all three dimensions. The checked official `main` still has the unguarded path. A test should run past a would-be event with pairs disabled and confirm that the run completes without creating a pair.

8. `identify` silently replaces charge or mass

In an EPOCH species block, `identify` can replace an explicit `charge` or `mass` without warning. A deck may then run with different particle properties than its author expected. The fork commit `b908c8591` adds a warning in all three dimensions. It does not change which properties EPOCH uses. The warning is absent from the checked official `main`. Tests should cover both input orders and both matching and mismatching values.

4 Improvements

Improvement list

These entries describe changes to a method or feature. The status column shows where each improvement is available; it does not claim that a missing improvement is a bug.

ID	Improvement	Available in	Credits
I01	Numerical nonlinear Breit–Wheeler rate table	**Fork**	Zheyuan Chen
I02	Full Landau–Lifshitz force and midpoint radiation-reaction step	**Fork**	Zheyuan Chen
I03	Load laser profiles and phases from data files	**Fork**	Zheyuan Chen
I04	Spin and photon polarisation in EPOCH	**Fork**	Holger Schmitz; Zheyuan Chen

I01. Numerical nonlinear Breit–Wheeler rate table

Nonlinear Breit–Wheeler (NBW) pair creation is the conversion of a high-energy photon into an electron–positron pair in a strong electromagnetic field. Official 2D EPOCH uses an approximate analytical rate in `pairprod.table`. The 2D fork adds a numerical table for the locally constant field approximation (LCFA), `pairprod_lcfa.table` (`848d21a4b`), selected through `pair_tables = lcfa` (`fd56a2e08`). The old method remains available as `legacy`. This is an improvement candidate, not a claim that the old method is a coding error. It can change simulation results, so record the chosen table and its hash. The local table generator is currently not committed to the fork; clean regeneration should be documented before proposing this for official EPOCH.

I02. Fuller radiation-reaction model and time step

The 2D fork adds optional terms of the full Landau–Lifshitz radiation-reaction force, another Gaunt-factor fit, and a midpoint time step for the radiation-reaction force (`dfa48930a`). The old options remain the defaults. These additions are not in the checked official `main`; the audited fork branch has no equivalent 3D version.

I03. Laser profiles and phases from data files

Zheyuan Chen's fork can read a laser's spatial or time-dependent profile from a data file in 2D and 3D (`438424c31`). The laser block accepts `use_custom_profile` and `profile_data_file`; the file-based phase option uses `use_phase_from_file` and `phase_data_file`. The implementation also supports separate data for more than one laser. This feature is absent from the checked official `main`. A portable example should supply the data file, its format and grid convention, and a reference field trace; a multi-rank test should also check file loading.

I04. Spin and photon polarisation

The fork's `spin` branch adds particle spin and photon polarisation tracking, with spin-dependent strong-field QED processes in 2D. Zheyuan Chen began this work by adapting Holger Schmitz's spin-precession code (`ea37e1a6d`); subsequent fork work adds spin-dependent emission and pair production (`773c614c3`, `2b18ef6b4`). The checked official `main` has no equivalent spin model. The 3D spin/QED port in the fork should be treated as experimental until its validation is documented. This entry describes the feature as a whole; it is not a registry of bugs found only within that feature.